MOFCOM-UNDP
Trilateral cooperation on Renewable Energy in Ethiopia

Revised on 4 March 2019

Sector: Renewable Energy
South-South Cooperation Location: China – Ethiopia – Sri Lanka
China Executing Agency: United Nations Development Programme China
Ethiopia Executing Agency: United Nations Development Programme Ethiopia
Implementing Partners: Ethiopia Ministry of Water, Irrigation and Electricity/China Ministry of Science and Technology/China Agricultural University
Project Duration: 3 Years
Project Budget: 2.042.394 USD

United Nations Development Programme

Partner Countries: China and Ethiopia

South-South & Triangular Cooperation Project Document

Project Title: Trilateral Cooperation on Biogas and Solar - Transitioning to Sustainable Energy Uses in public and private institutions including in the Agro-Industry in Sri Lanka - China – Ethiopia.

UNDAF/CPD Outcomes:

China Outcome 3: China expands and improves the effectiveness of its development cooperation with other developing countries through TSSC framework

Ethiopia Outcome 5: By 2020 key Government institutions at federal and regional levels including cities are better able to plan, implement and monitor priority climate change mitigation and adaptation actions and sustainable resource management.

Expected Output(s):

China UNDAF: Output 3. The effectiveness of China's engagement in international cooperation is enhanced for the mutual benefit of China and the world.

Indicator:

Ethiopia UNDAF Output 4.3: Technologies and practices including finance and market mechanisms that promote a climate resilient green economy introduced and scaled up

Implementing Partners:

Ministry of Water, Irrigation and Electricity

China Agricultural University

Ministry of Science and Technology, Administrative Centre for China's Agenda 21

Brief Description

The Project establishes a mechanism for trilateral and South-South Cooperation to address sustainable development challenges related to international cooperation, access to energy for improved service delivery and sustainable integrated farming practices. The Project takes a point of departure in the need for renewable energy technology dissemination and scale-up for Ethiopia climate resilient growth, and supports access to energy and sustainable resources consumption through trial and demonstration of biogas and solar energy for productive uses. The Project is implemented under a South-South Cooperation framework and as such presents itself as a learning platform for both China and Ethiopia to engage and cooperate at the international level in renewable energy technology and skills transfer.

Project interventions include:

- Developing capacity of stakeholders in the energy sector to assess the potential of RET in meeting energy needs in productive sectors such as public service delivery and the agro-industry.
- Sharing Chinese knowledge and experience in biogas and solar energy for productive uses.
- Determining suitable business models through South-South knowledge and experience sharing platforms.
- Providing support to Chinese institutions on carrying out relevant capacity development in South partner countries.

Total resources required 2,042,394 USD

Total allocated resources: 2,042,394 USD

- Funding from Ministry of Commerce of People's Republic of China: 1,000,000 USD
- Funding from Ethiopia Ministry of Water, Irrigation and Electricity and UNDP: 1,042,394 USD

Programme Period
(In respective participating countries): May 2019-May 2022___

Atlas Award ID: _____

Start date: __May 2019___
End Date ___May_2022____

PAC Meeting Date _____

Management Arrangements _____
(In respective participating countries)

Agreed by *China MoFCOM* _____ 1 April, 2019

Agreed by *Ethiopia MoFEC:* _____

Agreed by Implementing Partner: Ethiopia MOWIE _____

Agreed by China Executing Agency: UNDP China _____

Agreed by Ethiopia Executing Agency: UNDP Ethiopia _____

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List of Abbreviations

AETDPD	Alternative Energy Technology Development and Promotion Directorate
AEMFI	Association of Ethiopian Microfinance Institutions
CD	Capacity Development
CCS	Climate Change Secretariat
CSO	Community Service Organization
DBE	Development Bank of Ethiopia
ECIC	Ethiopia Climate Innovation Centre
EE	Energy Efficiency
EEA	Ethiopian Economic Association
ESA	Ethiopian Standard Agency
ESCOs	Energy Service Companies
ESD	Energy Services Directorate
FeMSEDA	Federal Micro and Small Enterprises Development Agency
FSPs	Financial Service Providers
GEF	Global Environment Facility
GEPU	Greening Ethiopian Public Universities initiative
GHG	Greenhouse Gas
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit
GoC	Government of China
GoE	Government of Ethiopia
HOAREC	Horn of Africa Regional Environment Centre
ICS	Improve Cook Stoves
M&E	Monitoring and Evaluation
MFIs	Microfinance Institutions
MOPRE	Sri Lanka Ministry of Power and Renewable Energy
MOST	Ministry of Science and Technology
MOWIE	Ethiopian Ministry of Water, Irrigation and Energy
MRV	Monitoring, Reporting and Verification
MSMEs	Micro, Small and Medium Enterprises
NBPE	National Biogas Programme Ethiopia
PFIs	Public Financial Institutions
PMU	Project Management Unit
PSC	Project Steering Committee
PV	Photovoltaic
REF	Rural Electrification Fund
REMP	Rural Electrification Master Plan
RETs	Renewable Energy Technologies
RFP	Request for Proposal
SFM	Sustainable finance Mechanism
SHS	Solar Home System
SNV	Netherlands Development Association
SLA	Service Level Agreement
TA	Technical Assistance
TSCC	Trilateral South-South Cooperation
UNCDF	United Nations Capital Development Fund
UNDP	United Nations Development Program
USAID	United States Agency for International Development
WB	World Bank

I. Situation Analysis

South-South Cooperation is a broad framework for collaboration among countries of the South within the political, economic, social, cultural, environmental, and technical domains. South-South Cooperation involves two or more developing countries, multi-lateral organizations or partners, and can take place on a bilateral, regional, sub-regional or interregional basis. This project is designed as a South-South Cooperation project to be implemented through a trilateral cooperation modality.

UNDP emphasizes the importance of South-South Cooperation in its Strategic Plan as a welcomed contribution to global development and traditional aid flows. Trilateral South-South Cooperation is a unique approach in which three parties cooperate to implement innovative solutions. Projects are designed as dynamic platforms for exchange and partnership between developing countries. The underlying principle is that each country benefits from transfer of specific technical capacities, knowledge exchange and policy development.

UNDP's role in Trilateral South-South Cooperation projects is to coordinate the exchange of knowledge and experience. UNDP also acts as a convener of international cooperation best practices in project formulation (i.e. integrating beneficiary country needs and development priorities, project partners' ownership) and implementation (i.e. following standard project management and reporting procedures, ensuring quality capacity development activities, and partners' equality in cooperation modalities).

Trilateral South-South Cooperation have similar focus areas, including:

- Sharing knowledge and experience;
- Addressing country development challenges by leveraging Chinese experience and technical know-how;
- Increasing cooperation opportunities;
- A demand driven process.

In March 2018, China undertook initial steps towards the establishment of the China International Development Cooperation Administration (CIDCA). CIDCA will be responsible for making strategic guidelines, plans and policies on foreign aid as well as coordinating and making suggestions on major issues related to it. CIDCA also oversees the allocation of aid and the evaluation of its implementation.

In line with UNDP's emphasis and prioritization of South-South Cooperation, CIDCA marks a reinforcement of China's commitment to strengthen knowledge exchange and experience in the global south. Particularly the Belt and Road Initiative (BRI), to which Sri Lanka and Ethiopia are partner countries, will be a key priority for CIDCA and directly linked to UNDPs existing efforts to support the success of BRI

In December 2015, a call for proposals for new Trilateral South-South Cooperation projects was initiated. The call for proposals focused on three thematic areas: sustainable energy solutions; livelihood creation, tourism development and wildlife protection; and disaster risk reduction. Both Sri Lanka and Ethiopia project proposals were selected. In both countries, the Trilateral South-South Cooperation initiative builds upon existing GEF-UNDP engagements on renewable energy technology adoption and dissemination, and government development priorities. While the projects have been developed as stand-alone projects, they share common aspects such as the same Chinese implementing partners, common activities, and management modalities.

The interventions detailed in this document focus on the Ethiopia Trilateral South-South Cooperation project.

Socio-Economic Background

Ethiopia is a low-income country (US$590 GNI/capita in 2015) with 100 million people, 80 percent of whom live in rural areas. Agriculture is the principal economic activity for three-quarter of the population, and it is responsible for 40% of total domestic production. Agriculture in Ethiopia is principally based on small holdings for both crop and livestock producers. Industry and services are growing activities and now account for 14% and 46% of GDP respectively.

Ethiopia has made significant progress in increasing access to basic services in both urban and rural areas to reach the MDGs. Access to education, health services, and water supply has been a Government priority and 40 000 schools, 40 universities and colleges, 20 000 health centres, 400 hospitals, and more than 100 000 rural water supply systems are providing basic services to the population.

Energy Access for Sustainable Development

The lack of access to modern energy sources in Ethiopia remains an important challenge and barrier to socio-economic development in terms of:

- Sustainability of resource consumption;
- Power for lighting, cooking, improved public service delivery and income generation.

Ninety percent of the total energy consumed is derived from primary biomass (i.e. wood fuel) and mainly consumed by the household sector for cooking and baking. Practically all the rural population and a significant proportion in urban areas uses solid biofuels (wood fuel, charcoal) for cooking and baking. Such heavy reliance on solid biomass fuels, the lack of penetration of efficient cook stoves and alternatives for biofuel production have impacts on the local environment and people's health. Emissions from forest degradation alone contributes to 50 million CO_{2e} and is the largest contributor of GHG emissions from the energy sector in Ethiopia. Furthermore, households are exposed to indoor air pollution from cooking fires.

Access to electricity is limited and unreliable. Only 25% of the population, and fewer than 5% of the rural population is connected to the power grid. Considering the slow pace of grid connection and continued population growth, it is estimated that even with an aggressive electricity distribution plan, only 50% of the population will have access to electricity by 2025. In grid area, power blackouts have increased in frequency and length. Power cuts usually last between 4 to 6 hours.

While energy plans are progressively addressing these challenges, the energy sector remains driven by a centralized model with priority to increase the generation capacity through large renewable energy systems (e.g. hydropower, wind and geothermal) and grid extension. Past projects to increase planning capacity to deliver distributed energy solutions in off-grid areas have not been as successful as expected. The Rural Electrification Master Plan (REMP) has not been updated since 2006, and the role of the Rural Electrification Fund (REF) for off-grid projects implementation has apparently not been institutionalized.

In this situation, the Government is relying on the private sector to deliver adapted energy solutions (i.e. lighting devices, solar home systems, solar water heaters), and promote clean and efficient cooking energy systems (i.e. improved cook stoves, biomass briquettes, sustainably-produced charcoal and domestic biogas plants).

Ongoing UNDP-GEF Project

To support the Government of Ethiopia, UNDP is currently implementing a GEF project to: "promote and encourage significantly greater use of renewable energy technologies for household and productive uses in rural communities in Ethiopia". The Project specifically targets market driven dissemination of small-scale lighting and cooking solutions to low income households through:

- Market stimulation, by generating access to micro-finance for the purchase of small-scale RETs and raising awareness on RETs benefits;
- Facilitation of the supply chain through technical standards definition and enforcement as well as capacity development support to Micro, Small and Medium Enterprises (MSMEs).

Implementation is planned in rural areas of the nine regional states, and will be led by Regional Energy Bureaus.

The GEF project has been designed through a bottom-up approach. Affordability and scattered population settlement are major issues for RETs market development. As a result, the GEF project has targeted technologies that appear the most adapted to meet households' energy needs and cost-efficient for GHG emission reduction (i.e. solar lanterns, Pico photovoltaic system, Improved Cook Stoves). Energy for productive uses has mainly been considered through the provision of improved lighting and phone charging to small service and commercial businesses, and improved cook stoves to baking shops and small restaurants. However, energy use in public institutions, and for agricultural and farming activities may require different technology answers.

The TSCC provides technological answers targeting larger energy needs for improved service delivery and increased sustainable energy uses in agriculture and farming.

Targeting Energy for Productive Uses

Besides the GEF project, several development partners are involved in the dissemination of renewable energy technologies in Ethiopia (See Annex 2 p.37). Most of these initiatives are targeting off-grid and households' energy uses. The TSSC project focuses on medium and large technological systems to target renewable energy adoption for productive uses.

Improved public service delivery. Most rural public institutions (i.e. schools, health posts and centres, water supply points) do not have access to electricity which limits the extent and quality of their services.

- 20,000 off-grid rural schools
- 15,000 off-grid rural health posts and centres
- 30,000 off-grid rural water supply points

Energy access for productive activities. Crop cultivation, farming and agro-processing are also constrained because of the limited reach of the power grid in rural areas. Thus, they depend on unreliable and expensive fuel-based systems as well as biomass. There are however, thousands of off-grid enterprises with on-site resources for self-generation of energy, incl. livestock farms, slaughter houses, and coffee processors.

Sustainability. Public institutions are a major consumer of fuelwood, which contributes to forest degradation. Public institutions such as universities cater for thousands of students, and still rely mainly on fuelwood for cooking and baking.

While improving access to, and reliability of energy services, renewable energy systems if adapted to the needs and well managed, may also reduce energy costs, increase income generation opportunities, and reduce unsustainable consumption of local resources.

Gaps and Challenges

Although plans to directly address energy productive uses have not been formulated, demonstration systems have been installed in both on-grid and off-grid areas including SPV systems in health centres, schools, universities and for community water supply as well as biogas systems in universities and agro industries. However, many of the installed systems have been abandoned or are not operating properly. Lessons learnt from these projects highlight various reasons such as:

- The lack of financial incentives and access to finance for investment;
- Inappropriate management systems;
- System technical failure attributed to lack of technical capacities in system design, construction, operation and maintenance;
- Non-availability of spare parts and quality equipment;
- Systems are not meeting energy needs;
- Lack of coordination between energy stakeholders (government agencies, private sector, and academic researchers);
- Lack of awareness on technology potential, especially with regards to biogas;
- Lack of policy and regulatory framework.

Solar and Biogas Energy in China's Agriculture Sector

Since late 1970s and early 1980s China started using biogas technology as an element in its agriculture modernization programs. By the end of 2006, the total annual biogas production reached about 8.5 billion cubic meters (incl. domestic biogas production), and more than 5,200 large and mid-sized biogas projects had been installed in livestock and poultry farms. China also developed 2,200 power grid biogas engineering projects for waste management in intensive animal farms, treating more than 60 million tons of manure a year.

To facilitate the usage of biogas the government has developed biogas technical training courses in Shanxi Province, and in 2005 trained and certified more than 4,000 biogas professional technicians. The Ministry of Agriculture which administers the Chengdu Biogas Scientific Research Institute (BIOMA) also operates an international training programme. The China Agricultural University is conducting R&D on biogas related technology; fermentation techniques; utilization of biogas and slurry; feedstock material and construction; development of biogas appliances.

A strong government commitment has resulted in the rapid development of the biogas technology in China and private sector expansion, with 2,810 biogas companies engaged in construction of biogas plants and 1,914 companies specialized in the production of biogas appliances.

Solar technology is expanding in China, and China now accounts for nearly two-thirds of the entire world's production of solar photovoltaics. For years, China has supported access to solar technology programs in developing countries, including solar PV, solar water pumps, solar thermal and solar cook stoves.

Trilateral South-South Cooperation between China, UNDP and Ethiopia: Innovation and Opportunities

Some large manufacturers of solar photovoltaic systems (SPV) have already built long term partnerships with energy service providers in Addis Ababa. They supply SPV and solar lanterns parts to local

manufacturers. At the same time, Chinese traders flood the local market with cheap but unreliable solar technologies.

With a proven track record of technology dissemination to meet various needs for its own development, supported by successful policies and business models, China has much more to offer in the field of renewable energy. The South-South cooperation framework is an opportunity to:

- Leverage China's experience in biogas and solar to address Ethiopia's energy needs for productive uses,
- Coordinate and drive China's know-how and experience towards national development goals in Ethiopia.

The innovation in the TSSC project approach lies in the organizational setting of the project. Various stakeholders are brought together under the umbrella of a unique project, addressing multiple development challenges and government priorities in both countries.

This project utilizes UNDP's global experience in encouraging climate-resilient economic development. UNDP has a large environmental portfolio globally and has accumulated abundant experience in tackling climate change mainly through carrying out GEF projects. Based on China's expertise in renewable energy transfer and climate change mitigation, this project will foster exchanges and cooperation between China and Ethiopia, and support the adoption and appropriate uses of RETs for:

- Increased access to modern energy in off-grid areas for productive uses;
- Improved delivery of public services in on/off grid areas;
- The emergence of sustainable, integrated agriculture and farming practices for increased income generation and employment opportunities, as well as GHG emission reduction.

The project supports China's new policy on Climate Change South-South Cooperation as well as Ethiopia's Climate Resilient Green Economy Policy through adoption and dissemination of renewable energy technologies. The TSSC has been formulated with a focus on priority areas identified by Ethiopia in light of China's capacity to provide support to the renewable energy sector. By engaging expertise, knowledge and technology from multiple Chinese entities, the project also increases coordination and coherence among Chinese government agencies in delivering assistance.

Through its capacity development activities, the project also supports Chinese institutions in carrying out relevant capacity development in south partner countries.

II. Strategy

Theory of Change

The theory of change articulates the multiple interventions required to reach a long-term objective. Activities are planned along a causal pathway where they nurture a set of preconditions (outputs/outcomes) to be met to achieve the objective. Based on identified challenges presented in the situation analysis, the Government of Ethiopia, the Government of China and UNDP have defined the following project objective:

To build a successful case for medium scale[1] biogas and integrated energy system through demonstration projects for productive uses.

The preconditions for meeting the objective include:

1. Developing capacity of stakeholders in the energy sector to assess the potential of RET to support Ethiopia climate resilient growth with a focus on:
   - Access to modern energy in off-grid areas for productive uses;
   - Improvement of public services delivery in on/off grid areas;
   - Emergence of sustainable, integrated agriculture and farming practices for increased income generation and employment opportunities as well as GHG emission reduction.

2. Sharing Chinese knowledge and experience in biogas and solar for productive uses.

3. Determining suitable business models through South-South knowledge and experience sharing platforms.

4. Providing support to Chinese institutions on carrying out relevant capacity development in South partner countries.

Rationale and Approach

Leveraging support and experience from China

With a proven track record of renewable energy technology dissemination to meet various needs for its own development, supported by successful policies and business models, China is a strategic partner for the project. The project relies on technical know-how from The Ministry of Science and Technology Administrative Centre for China's Agenda 21 (ACCA21) and the China Agricultural University (CAU).

The Ministry of Science and Technology Administrative Centre for China's Agenda 21 (ACCA21)

Under the on-going South-South project for RET transfer in Ghana and Zambia, ACCA21 has developed:

- A large network of technology suppliers which can be used to link private sector actors between the three countries;
- Capacity to develop and manage a knowledge and experience exchange platform through the South-South Cooperation Centre (SSCC).

[1] Medium scale biogas plants in Ethiopia are defined as plants with a capacity comprised between 40 and 100/120 m^3.

ACCA21 experience will be used to develop suitable business models and facilitate partnerships with the private sector to reinforce the energy service industry in Ethiopia. ACCA21 will also use its existing network to institutionalize a platform of communication and exchanges to facilitate access to technical and financial resources. The overall aim is to provide access to technical information (E-learning course, technology manuals, etc.), and potential business partners in China to stimulate local businesses.

The China Agricultural University: CAU experience in R&D on biogas related technology; fermentation techniques; utilization of biogas and slurry; feedstock material and construction; development of biogas appliances essential to support capacity development activities and develop training material on biogas systems. CAU tested models on technology adaptation for the modernization of the agriculture sector (energy production, waste management, and integrated agricultural practices) will provide Ethiopia with a set of experiences and expertise to support its own transition. CAU will provide guidance to the energy needs assessment and ensure that technological answers are adapted to energy needs. It will also be a partner in facilitating joint research studies between Sri Lanka, Ethiopia and China.

Capacity development as an integrated project component

Capacity development is an integrated and fundamental aspect of the Trilateral South-South Cooperation framework. UNDP's approach to capacity development is to ensure that core principles of cooperation: national ownership, adaptation to local conditions, partners' equality, and use of national systems are embraced. UNDP's five step methodology to ensure sustainable capacity development outcomes has been translated into sub-activities within each capacity development component of the TSSC project. These steps are: 1) Engage stakeholders on capacity development; 2) Assess capacity and needs; 3) Formulate a capacity development response; 4) Implement the response; 5) Evaluate capacity development.

To ensure that learning takes place in the project/activity implementation process, stakeholders in Ethiopia and China will implement activities jointly. To that effect, some capacity development activities will be generated on-site, through the demonstration of Chinese technology and experience.

Renewable energy specialists from the China Agricultural University will assist with energy needs assessments, through a series of training and on-site mentoring/coaching activities. They will also ensure that, in the process of designing and constructing the generation facilities, local energy stakeholders (i.e. Government officials, local authorities, local association, and end-users) are effectively taught basic principles for construction, operation and maintenance of systems.

On-sites activities during the energy assessment and the construction phases will provide CAU experts with a field experience and knowledge of the Ethiopian local context (i.e. capacities, barriers and opportunities). The overall UNDP framework for capacity development and field activities will support Chinese institutions in carrying out relevant capacity development in south partner countries.

Stakeholder coordination

A key component of the project includes coordinating stakeholders' activities to generate lessons learnt. Knowledge transfer between project partners is critical to the project success. It will occur through collaborative activities encouraging information sharing, knowledge exchange platforms and development of an E-Learning course.

With UNDP support, MOWIE shall coordinate and convene stakeholders on a regular basis to get timely and relevant input to, for example, capacity development, generation of knowledge products and technology transfer.

A fundamental aspect of south-south cooperation between China and Ethiopia is to create linkages between research institutes, private sector companies and the government counterparts which help ensure long term sustainability. Mutual understanding and creating deep collaboration between Ethiopia and China can be ensured through UNDP facilitation, study visits, matchmaking between companies and establishing communication platforms (online/ offline). Communication platforms will not only involve the governments of both countries, but also enable energy service providers, financial institutions, and end-users to communicate efficiently.

Thematic alignment to national priorities

The project is aligned with UNDAF China Outcome 3 to expand and improve the effectiveness of China's development cooperation with other developing countries through TSSC framework; and UNDAF Ethiopia Outcome 5 targeting by 2020 that key Government institutions at federal and regional levels including cities are better able to plan, implement and monitor priority climate change mitigation and adaptation actions and sustainable resource management.

The project is also supporting national priorities in terms of increased renewable energy and sustainable resources consumption.

The off-grid electrification plan for the 2015-2020 period targets the distribution of 4 million solar home systems, and the installation of 355 renewable energy mini grids powered by solar and hydropower. The same plan includes electrification of 3600 public institutions (health posts and schools) with solar systems.

The plan for improving sustainability is focused on dissemination of clean cook stoves in the household (domestic) sector. During the 2015-2020 period the plan envisages distribution of 11 million clean wood stoves and 31,000 domestic biogas plants.

Finally, the project supports the Government of Ethiopia Climate Resilient Green Economy (CRGE) vision by introducing and scaling-up technologies that promote a climate resilient green economy. A major objective of the CRGE is to improve crop and livestock production practices for food security, and increased income generation while reducing GHG emissions. The technologies and cooperation modalities introduced in the TSSC project have the potential to contribute to:

- GoE low cost irrigation scheme and increase the uses of organic fertilizers and manure for improved crops yields and land management.
- Improved management of agriculture waste disposal for GHG emission reduction and environmental preservation.

Implementing Partners

The Ethiopian Ministry of Water, Irrigation and Electricity (MOWIE): is a federal organization established to undertake the management of the country's water resources, medium and large-scale irrigation and electricity resources. The Ministry is a regulatory body and provides technical support to regional water and energy bureaus.

The Energy Study and Development Follow-up Directorate (ESD): Under MOWIE the directorate is responsible for energy planning, development of guidelines, strategies, polices, and programs, and conducting study and research activities on energy

Alternative Energy Technology Development and Promotion Directorate (AETDPD): Under MOWIE the directorate is the implementing agency for renewable energy technology projects, responsible to:

- Increase access to affordable modern energy means;
- Develop RET thorough adoption, adaptation and new design;
- Create awareness on energy related matters using different promotion medias;
- Provide technical assistance in the form of advisory services, facilitating revolving fund, providing Training of Trainers (ToT).

The AETDP directorate is the actual implementing partner for the on-going GEF project.

The Ministry of Science and Technology, Administrative Centre for China's Agenda 21 (ACCA21): was established in 1994 to effectively promote the implementation of China's Agenda 21. ACCA21 mission is to support technological innovation and promote sustainable development in China. ACCA21 acts as an excellence research centre and project management unit for various projects. ACCA21 is also leading the creation of a South-South Cooperation Centre, which aim to become a platform of exchanges to facilitate and institutionalize partnerships with the South.

The China Agricultural University:

Under ***the China Agricultural University***, the Key Laboratory of Clean Production and Utilization of Renewable Energy focuses on the following research fields: pre-treatment of mixed-culture feedstock, advanced anaerobic digestion technology and equipment development, biogas clean production and safety discharge, technology evaluation and development strategy. The laboratory aims at promoting renewable energy development and providing consulting, research, development, and personnel training to developing countries.

Outcomes, Outputs and Activity Overview

Outcome 1: Capacity for RET transfer built through demonstration of biogas and solar for productive uses

The project seeks to support energy stakeholder's capacity development in planning and implementing efficient and adapted energy solutions. Biogas and solar are not new technologies to Ethiopia. However, the adoption of these technologies to meet energy needs for productive uses in on and off-grid areas has been limited. The following activities are expected to provide direction and support in the management, design, and implementation of these energy systems.

Chinese experts will assist MOWIE with this component. Responsible for capacity development and engaged in the energy needs assessment, the specialists' presence will also ensure Chinese exposure to Ethiopia local context (capacities, barriers, opportunities) at an early stage of the project.

Output 1.1: Key partners identified and demonstration sites selected

Energy planning efforts, still driven by a centralized approach of power grid expansion, have failed to consider the specificity of energy demand by type of use and socio-economic objective to be reached. As a result, energy projects are developed on an ad hoc basis by different stakeholders (i.e. NGOs, donors, electricity utility, government agencies, and researchers) and remain disconnected from an integrated and evidence-based decision-making process.

The activities will focus on strengthening MOWIE capacity to manage the transition to RET for sustainable energy use in public institutions and agro-industrial activities: i.e. assess needs, prioritize actions, and coordinate institutional set-up of RE projects.

Activity 1.1.1: Shortlisting potential institutions

To ensure high level management commitment to installation, cost of operation and maintenance of the system, MOWIE will send a communication to institutions, line ministries, associations (industry-academic association) with information on the project objectives and the plan to conduct energy need assessments.

The communication needs to address productive uses not only in the sense of powering facilities (lighting, heating, etc.), but also in terms of goods and services creation/improvement such as:

- Education;
- Health;
- Employment;
- Sustainable uses of resources;
- Income generation, etc.

MOWIE will call for a detailed request of interest, with an outline of roles, responsibilities, and expectations from interested institutions.

A long list of potential institutions/sites will then be established to conduct energy need assessments.

Activity 1.1.2: Undertake ten initial energy needs assessments

The objective of the energy needs assessment is to determine when renewables are appropriate options, and how does one select between technologies to meet the project objectives of:

- Increased access to modern energy in off-grid areas for productive uses
- Improved delivery of public services in on/off grid areas
- Support the emergence of sustainable, integrated agriculture and farming practices for increased income generation and employment opportunities as well as GHG emission reduction.

Based on the EOI, MOWIE will conduct up to ten energy needs assessment studies together with key partners including: Chinese experts/counterpart; solar associations; national biogas program experts, students from energy education programs (Addis Ababa Institute of Technology, MOST Centre of Excellence, etc.).

The energy needs assessment shall include social-, environmental- and economic-impact assessment to provide an overall understanding of the technologies' potential to achieve the project goals as outlined above. Technology selection will be conducted based on needs and potential socio-economic impacts

A steering committee will screen the needs assessments and recommend 3-5 optimal sites. Ideally, a diverse pool of institutions and associations will be selected including: on and off-grid university, school, hospital, TVET centre, orphanage, farm, and agriculture cooperatives.

Output 1.2: Capacity built through demonstration

The procurement will be managed by UNDP and should be developed to build a consortium:

- Providing optimal integrated RET solutions, based on the energy need assessments.
- Partnering with Ethiopian companies to ensure on-site training of technicians for operation and maintenance.
- Engaged in developing course curricula to train key stakeholders.

The selected consortium will be responsible for overseeing the complete planning and construction process according to defined rules and regulations.

Activity 1.2.1: Site planning and construction

Detailed feasibility study will be conducted for the 3-5 selected demonstration sites. They will include system costs and address the following:

- Design RET systems in collaboration with MOWIE and end-users (incl. detailed institutional arrangements for system ownership and operations);
- Detailed provision of operation and maintenance costs and management plan (incl. provision and replacement of spare parts and system hardware, existing suppliers, etc.);
- Conduct social and environmental screening;
- Development of a comprehensive monitoring system.

Activity 1.2.2: On-site training for operation and maintenance

Operation and maintenance are critical to ensure long term uses and benefits of the technology. Past experiences have demonstrated that without proper operation and maintenance plans, technology systems are abandoned. The Chinese implementing partner will therefore engage in on-site training for system operation and maintenance.

Curriculum, training manual and training will be provided to the 3-5 institutions operating the system and other key stakeholders for on the job-training of trainers, build capacity of private sector, RE associations, and technicians.

Outcome 2: South-South knowledge and experience exchanged

A number of approaches to sharing experience and knowledge will be developed. Initial ideas include an online platform where project knowledge products and training curricula are shared. In addition, offline hands-on workshops shall be conducted to share key lessons learned.

Output 2.1: Tailored mentorship/traineeship program for exchange with China developed and participants trained

The training will target two main needs. On one hand, developing a pool of experts capable to train in systems installation, operation and maintenance. On the other hand, energy service providers and MOWIE staff will be supported in developing capacities to design and plan biogas and solar systems.

For the success of biogas operation, technicians need to be trained in the anaerobic digestion process, both to understand the biological process (theory) as well as to maintain the right conditions for the system (practice). To ensure that technicians are exposed to both theory and practice, the project will support MOWIE in identifying potential candidates for MOFCOM trainings and scholarships. The trainings will be financed separately by MOFCOM. Coordination of support provided by the TSSC project and by parallel interventions from MOFCOM is critical for the success of this activity. During the inception phase, project partners will consult with MOFOCM on how to best address coordination needs.

One fundamental aspect of this output is to strengthen Chinese institutions in delivering adapted capacity development and training activities to South partner countries. To that effect, the UNDP approach to capacity development will be used, with a focus on ownership and adaptiveness, project objective oriented, and efficiency. Capacity development and training activities will be systematically evaluated based on these

criteria. These evaluations will serve to generate lessons learnt to adapt UNDP's support to South-South Cooperation initiatives.

Activity 2.1.1: Training of trainers in installation, operation and maintenance

The training courses will lead to the formulation of manual for operation and maintenance of specific system; for example, maintaining ph levels, feedstock, screening safety protocols, operation of machinery for biogas systems. Infographic, video, reading materials, quiz, etc. that can be used on the demonstration sites will be developed.

Activity 2.1.2: Training on system design

The training courses will lead to curricula development on designing biogas and solar systems and advisory services for agro-industry (i.e. energy audits, sizing and system optimization, costing systems, biogas upgrading, etc.), incl. online courses.

Finally, to reinforce the South-South knowledge exchanges, the Chinese implementing partner (e.g. China Agricultural University) will adapt the training material to an E-Learning course that will be made accessible by ACCA21 on the South-South Cooperation Centre website.

Output 2.2: Energy stakeholders in China and Ethiopia are exposed to best business practices in the RE industry

Under this output experience and business models will be developed to cater to hybrid RET systems such as biogas and solar PV.

The knowledge platform will address direct barriers and challenges faced by energy stakeholders, such as:

- Existing barriers to biogas and solar dissemination and recommendation for scaling-up the initiative.
- Existing operational barriers to private sector engagement, through the development of business networks and knowledge sharing on proven business models.
- Procurement of systems and appliances through Chinese suppliers as well as supply and maintenance of solar PV systems.

Activity 2.2.1: Develop and trial Renewable Ethiopia Awards

To promote good management and competition between institutions, an Award Show will be organized to encourage sector players to commit to RET adoption. Success stories will be showcased by stakeholders including, public institutions, private associations, research organizations.

Categories potentially including: "best China-Africa cooperation", "Greenest Universities", "Energy Efficiency", "Reduction of GHG", "Innovative RET uses", etc. will be developed under an overall concept.

The award show shall be nationally hosted to reinforce and coordinate with existing initiatives such as the Greening Ethiopian Public Universities (GEPU) initiative.

Activity 2.2.2 Facilitate business best practices forum

The forum will address the need for mainstreaming renewable energy technologies as viable alternatives to grid expansion to meet energy productive uses needs. The forum should present best practices to key stakeholders. Knowledge products will include:

- Study report on business models;
- Lessons learnt from demonstration sites;
- Technology assessment;
- An Investment plan for biogas and solar for productive uses. This will take the form of a strategy for scaling up initiative (including financing requirements, ToRs or concept notes for new projects).

Activity 2.2.3 Coordinate business match-making platform for energy service providers

The match making platform will help identify suppliers and facilitate business relationships to match potential demand for RETs in Ethiopia. Chinese implementing partner will identify Ethiopian energy service providers' demand in terms of access to technology hardware and develop an online catalogue of demand-driven biogas and solar PV products/appliances and technologies for agro-industry.

The Chinese implementing partner will also organize and host a trade fair for RET technology with possible site visits for technology demonstration and business match-making.

Activity 2.2.4:Joint research projects

Joint research projects should include private companies, technical agencies and local/provincial universities, and academic institutions from the three countries. In-country visits are expected.

Through a set of collaborative research and consultations the project will mainstream biogas and solar potential for a transition to sustainable energy consumption and production patterns. Studies and consultation will consider:

- How to disseminate and scale up RET use at institutions;
- Institutional and management barriers to RET adoption in institutions;
- Study procurement modalities for RETs importation and transfer.

Joint research studies will be conducted through partnership/twinning with Ethiopian, Sri Lanka and Chinese research institutes.

UNDP Roles and Functions

UNDPs role and mandate in a trilateral project falls into the following priorities for United Nations support listed below:

1. Support South-South cooperation efforts;
2. Strengthen institutional and technical capacities;
3. Improve the exchange of knowledge, experience and know-how among China-Ethiopia and Sri Lanka;
4. Respond to the specific development challenges of partner countries;

The following table provides a summary of what and how UNDP can provide support to contribute to achieving expected outcomes:

Role (what)	How (processes)	Outcomes
Convener	• Bring parties together and facilitate dialogue • Support partners in identifying and planning project approaches and implementation	• Consensus reached and project formulated • Consultative and inclusive processes that take partners needs and concerns into account. • Technical capacities of Ethiopian energy stakeholders strengthened, including private sector • Experience on SSC institutionalized within key partners in China.
Knowledge broker	• Bridge knowledge gaps by linking supply and demand of expertise and experience • Identify, document and disseminate good practices • Ensure quality of training and capacity development activities	• Knowledge exchange platform between Ethiopia, China and Sri Lanka facilitated • Successful practices documented for adaptation or scaling up • Ethiopian institutions connected and networked • UNDP approach and methodology for capacity development used and adapted delivered. Capacity development evaluated
Partnership builder	• Rally partners and key project stakeholders • Organize forums, meeting and events • Mobilize resources • Coordinate and leverage UN and other donor support	• Inclusive partnerships and strategic alliances forged between Ethiopia, China and Sri Lanka • Technical and financial resources mobilized • MOFCOM and GEF co-funding is efficiently managed to support project objective achievements

UNDP will specifically provide project oversight through its project management setup, modalities and its standard rules and procedures. Project oversight ensures that objectives and outcomes are met and management is adaptive to the context in which the project operates. This includes assessing strategic opportunities as they arise and adopting these when relevant. UNDP will play an important role in facilitating communication and coordination between the implementing partners. At the activity level, UNDP will support processes of capacity development that takes place between the implementing partners and provide quality assurance services to that effect.

At the national strategy level, UNDP, through its roles and functions outlined above, will contribute to both China and Ethiopia national priorities and strategic outlook.

III. Joint Results and Resources Framework

Intended Outcomes as stated in the respective CPs' RRFs: *China Outcome 3: China expands and improves the effectiveness of its development cooperation with other developing countries through TSSC framework* *Ethiopia Outcome 5: By 2020 key Government institutions at federal and regional levels including cities are better able to plan, implement and monitor priority climate change mitigation and adaptation actions and sustainable resource management.*
Outcome indicators *(as stated in the RRF of CP of respective partner countries, including baselines and targets):* *Ethiopia UNDAF Output 4.3: Technologies and practices, including finance and market mechanisms, that promote a climate resilient green economy introduced and scaled up* *Indicator: Number of green technologies introduced at national level* *Baseline: No/only few number of functional large scale institutional biogas and solar plants* *Target: 4 new large scale institutional biogas and solar systems for demonstration* *China UNDAF: Output 3. The effectiveness of China's engagement in international cooperation is enhanced for the mutual benefit of China and the world.* *Indicator: Increased and effective engagement of China in south-south cooperation OR Number of South-South development cooperation between China and developing countries* *Baseline: Zero (2019)* *Target: 1 by 2020*
Partnership Strategy: *Ethiopia Ministry of Water, Irrigation and Electricity(MOWIE)- Administrative Centre for China's Agenda 21 (ACCA21) - China Agricultural University (CAU)*

Project title and ATLAS Project (formerly Award) ID: Biogas and Solar Trilateral Cooperation - Transitioning to Sustainable Energy Uses in the Agro-Industry in Sri Lanka - China – Ethiopia. Ethiopia Project								
Outcome 1: Capacity for RET transfer built through demonstration of biogas and solar for productive uses								
INTENDED OUTPUTS	OUTPUT TARGETS FOR (YEARS)	INDICATIVE ACTIVITIES	RESPONSIBLE PARTIES	Input Matrix				
				China	MOWIE	UNDP Ethiopia	UNDP China	Funding
Output 1.1: Enhanced partnerships among key partners through demonstration of biogas and solar for productive uses **Country/ Implementing Partner:** Ethiopia/MOWIE **Indicators:** - No of EOIs received - List of 10 sites for energy assessments conducted - No of partnerships established for conducting energy needs assessments - Roadmap for energy needs assessments prepared -No of energy needs assessments reports delivered - No of social and environmental impact assessments delivered - % of steering committee representatives endorsing 3-5 selected sites **Baselines:** none started /completed	Targets (year 1) MOWIE communication sent 10 sites for energy need assessments identified with end-users' commitment. 3 teams based on multi-stakeholders' partnerships selected to conduct energy need assessments Training on methodology and objectives conducted 10 energy need assessments conducted Short list of 3-5 recommended sites form the listed 10 sites	Activity 1.1.1: Shortlisting potential institutions Activity 1.1.2: Undertake initial energy needs assessments (in ten institutions)	*MOWIE* *MOWIE*	*Support from CAU for methodology* *China TA*	*Facilitation and Coordination*	*Overall QA*	*Overall QA*	*MOFCOM* *MOFCOM*

Output 1.2 Capacity built through RE demonstration site construction **Country/IP:** Ethiopia/MOWIE **Indicators:** - No of sites selected - No of feasibility studies endorsed - Equipment that meet quality and cost-efficiency standards procured as per the national standard - No of capacity need assessments - Training plan prepared -No of people trained - No of Operation, maintenance and safety manuals developed/updated -%of positive evaluation of training - No of operational RET systems at the end of the project **Baselines: none started / completed**	Targets (year 1) 3-5 feasibility studies delivered One EPC signed and RET systems procured Targets (year 2) 3-5 sites equipped Stakeholders engagement-capacity need assessment and training plan At least one staff per demonstration site/at least 1 staff per MOWIE&REB/ at least 1 staff from local association) trained in operation and maintenance At least 1 Operation, maintenance and safety manual delivered per site Monitoring of systems operation	Activity 1.2.1: Site planning and construction *Feasibility study* *Equipment* Activity 1.2.2 On-site training for operation and maintenance	*MOWIE* *UNDP China* *MOWIE*		*Facilitate and ensure partnership between local stakeholders and Chinese EPC staff for training Quality control of equipment*	*QA feasibility studies Supervise installation and training*	*QA feasibility studies Procurement*	*UNDP Ethiopia/GEF* *Co-financed MOFCOM/UNDP Ethiopia* *MOFCOM*

Outcome 2: South-South Knowledge and Experience Exchanged								
INTENDED OUTPUTS	OUTPUT TARGETS FOR (YEARS)	INDICATIVE ACTIVITIES	RESPONSIBLE PARTIES	Input Matrix				
				China	MOWIE	UNDP Ethiopia	UNDP China	Funding
Output 2.1: Tailored mentorship/traineeship program for exchange with China developed and participants trained	Target (Year 1) Consultation, capacity need assessment, identification of trainees Design training of trainers courses for construction,	Activity 2.1.1 Training of trainers in installation, operation and maintenance Activity 2.1.2: Training on system design	*CAU* *CAU*	*ACCA21 to coordinate and facilitate E-learning course*	*staff for training Coordinate with GEF business incubator*	*QA capacity development*	*QA capacity development*	*UNDP Ethiopia/GEF* *UNDP Ethiopia/GEF*

Country/IP: China/CAU - No of capacity need assessment reports - No of participants to training - Training course material - No of participants to system design training -% of positive evaluation of training - number of hands-on training course for design and optimization - number of eLearning course on SSC Centre website -No of views and subscription to online course Baselines: none started / completed	operation and maintenance Training of trainers courses for construction, operation and maintenance in China (At least 1 staff from MOWIE/1 staff from National Biogas Program/ 1 staff from business incubator program under GEF) Capacity need assessment in system design Design training courses on system design and optimization Training courses on system design and optimization held in Ethiopia Target (Year 2) - RE Training curricula added on SSC centre website as E-Learning course				*activities*			
Output 2.2 Energy Stakeholders in China and Ethiopia are exposed to best business practices in the RE industry **Country/IP:** China/ACCA21 **Indicators:** - No of reports produced and presented to best practices workshop - No of participants to best practices workshop	Target (Year 1) Conduct private energy service providers mapping in China (biogas and solar) Online Catalogue of suppliers/appliances developed and regularly updated Target (Year 2) Award designed and hosted Study report finalized and presented to best practices workshop in Ethiopia: - Study Report on business	Activity 2.2.1: Develop and trial Renewable Ethiopia Awards Activity 2.2.2: Facilitate Business Best Practices Forum Activity 2.2.3: Coordinate business match-making platform for energy service providers Activity 2.2.4: Joint Research	MOWIE ACCA21 ACCA21 CAU		*Coordinate with GEF activities*	Review list of participants to business forum, planned site visit and trade fair	Design Award	*UNDP Ethiopia/*GEF MOFCOM MOFCOM MOFCOM

- No of Ethiopia participants to business platform in China - % of participants satisfied with business match-making platform and trade fair - No of consultation of online catalogue - % of users satisfied with online catalogue service **Baselines: none**	models - Technology assessment report - Investment Plan for biogas and solar - Lessons Learnt from demonstration sites Financial stakeholders mapping in China & Ethiopia. At least 2 representatives of financial institutions involved in renewable energy portfolio invited to best practices workshop Best practices workshop with at least 50 people stakeholders held in Ethiopia At least 10 stakeholders from Ethiopia (private companies, biogas association, government counterparts) invited to workshop and business match-making platform Trade fair hosted/Site visits in China	Projects						
Budget for China (MOFCOM) *(core, non-core, to be mobilized)* **Budget for Ethiopia (Government)** *(core, non-core, to be mobilized)*	1, 000,000 1,000,000							

IV. Multi-Year Work Plan by Partner Country[2]

Country: Ethiopia **Implementing Partner: Ethiopia-China**

EXPECTED OUTPUTS[3] *And baseline, indicators including annual targets*	PLANNED ACTIVITIES *List activity results and associated actions*	TIMEFRAME			RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1	Y2	Y3		Funding Source	Budget Description	Amount
Outcome 1: Capacity for RET transfer built through demonstration of biogas and solar for productive uses								
Output 1.1: Enhanced partnerships among key partners through demonstration of biogas and solar for productive uses **Indicators:** - No of EOIs received - List of 10 sites for energy assessments conducted - No of partnerships established for conducting energy needs assessments - Roadmap for energy needs assessments prepared -No of energy needs assessments reports delivered - No of social and environmental impact assessments delivered - % of steering committee representatives endorsing 3-5 selected sites **Baselines:** none **Related CP outcome:** *Ethiopia 5*	Activity 1.1.1: Shortlisting potential institutions	X			MOWIE	MOFCOM		37,000
	Activity 1.1.2: Undertake 10 initial energy needs assessments	X			MOWIE	MOFCOM		132,000

[2] A separate, subsidiary multi-year AWP should be prepared **for each partner country** with an implementing partner. The AWPs for Year 1 can also be signed when the project document is signed to facilitate timely implementation.

[3] Multiple countries/IPs can contribute to the same output and can share the same indicators. **Only the contribution by one country/IP should be present in any one subsidiary country AWP.** Joint results are included in the joint RRF. All AWPs together will yield the joint RRF.

Output 1.2 Capacity built through RE demonstration site construction **Indicators:** - No of sites selected - No of feasibility studies endorsed - Equipment that meet quality and cost-efficiency standards procured as per the national standard - Training plan prepared -No of people trained - No of Operation, maintenance and safety manuals -%of positive evaluation of training - No of operational RET systems at the end of the project **Baselines: none** **Related CP outcome**: *Ethiopia 5/ China 3*	Activity 1.2.1: Site planning and construction		X		MOWIE	*UNDP Ethiopia*/GEF	Feasibility Studies	150,000
					UNDP China	*MOFCOM*	Equipment	600,000
					UNDP Ethiopia	UNDP Ethiopia	Equipment	300,000
							Other costs	37,000
	Activity 1.2.2 On-site training for operation and maintenance		X		MOWIE	MOFCOM		16,000
Outcome 2: South-South knowledge and Experience Exchanged								
Output 2.1: Tailored mentorship/traineeship program for exchange with China developed	Activity 2.1.1 Training of trainers in installation, operation and maintenance	X			CAU	*UNDP Ethiopia*/GEF		71,900

and participants trained **Indicators:** - No of capacity need assessment reports - No of participants to training - Training course material - No of participants to system design training - Hands-on training course for design and optimization -% of positive evaluation of training - E-Learning course on SSC Centre website -No of views and subscription to online course **Baselines:** none **Related CP outcome**: *Ethiopia 5/ China 3*	Activity 2.1.2: Training on system design	X			CAU	*UNDP Ethiopia*/GEF		78,900
Output 2.2 Energy Stakeholders in China and Ethiopia are exposed to best business practices in the RE industry **Indicators:** - No of reports produced and presented to best practices workshop - No of participants to best practices workshop - No of Ethiopia participants to business platform in China	Activity 2.2.1: Develop and trial Renewable Ethiopia Awards		X		MOWIE	*UNDP Ethiopia*/GEF		70,000
	Activity 2.2.2: Facilitate business Best Practices Forum		X		ACCA21	MOFCOM		91,000
	Activity 2.2.3: Coordinate business match-making platform for energy service providers				ACCA21	MOFCOM		67,300

- % of participants satisfied with business match-making platform and trade fair - No of consultation of online catalogue - % of users satisfied with online catalogue service **Baselines: none** **Related CP outcome**: *Ethiopia 5/ China 3*	Activity 2.2.4: Joint Research Projects				CAU/Ethiopia Universities and Academic networks	*MOFCOM*		15,000
Project Management	Project Identification and Stakeholder consultation	X						25,000
	Set-up project management structures in China	X						10,000
	Set-up project management and M& E structures in Ethiopia	X						76,000
	Inception/Launch workshop and study tour to Ethiopia	X						36,000
	PBM meeting and study tour in china	X						36,000
	Establish coordination mechanisms and communi-cation platform				UNDP China	MOFCOM		67,000
	Communication equipment				UNDP China	MOFCOM		4,000
GMS (8%)								122,094
Total								2,042,394

V. Management Arrangements

In China, the implementation modality will fall under UNDP's Direct Implementation Modality (DIM). UNDP China will be the executing agency on behalf of MOCOM and responsible for the management of Chinese implementing partner's technical and financial inputs to the TSSC project.

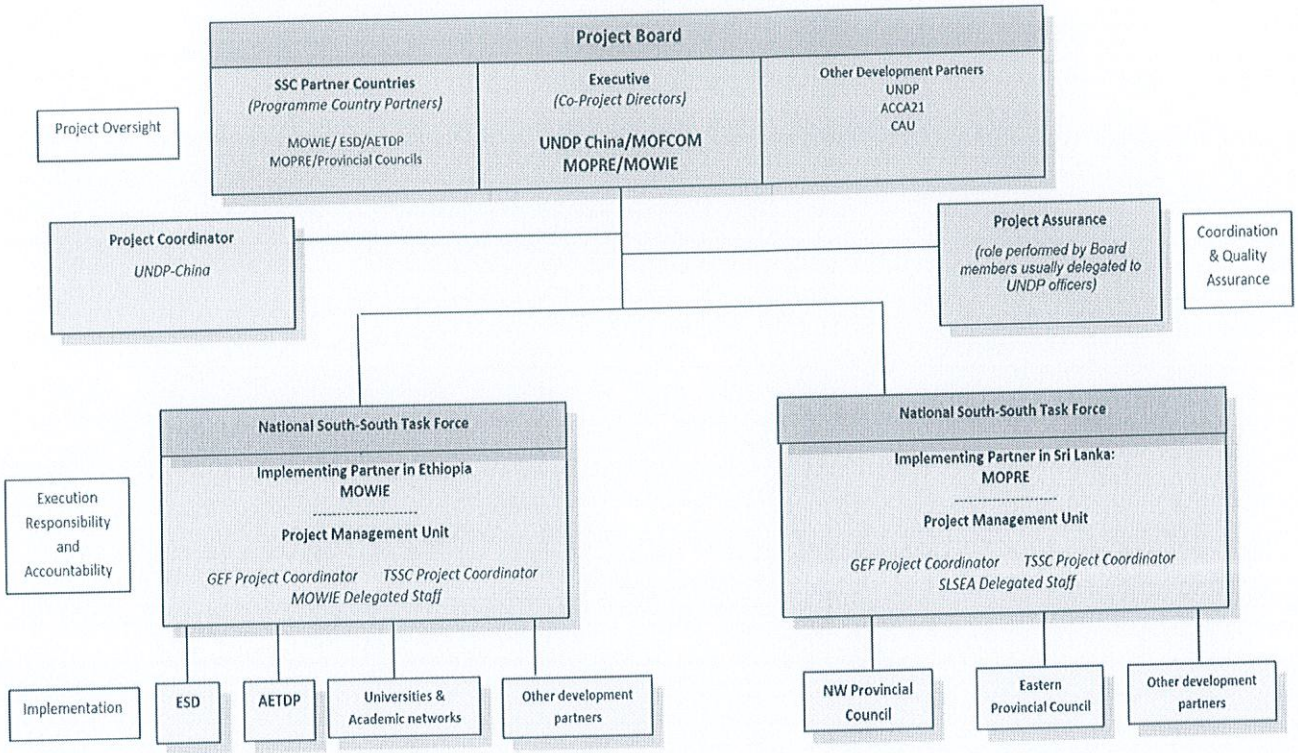
In Ethiopia, the implementation modality will fall under UNDP's Nationally Implemented Modality (NIM). The Ministry of Water, Irrigation and Electricity (MOWIE) will be both the project executing agency and the main Ethiopian implementing partner.

To facilitate a high degree of partnership and ownership among partners, who have not yet had the opportunity to meet and discuss in detail the organizational set-up, the inception phase includes a short consultation workshop. The objectives of the consultation workshop shall be:

- Confirm & adjust the organizational set-up;
- Confirm & adjust funding streams and accountability safeguards;
- Setup the project management units and procedures including elaboration of the programme/ project administration, communication, and coordination manual;
- Plan and agree upon the Annual Work Plan between Ethiopia and Chinese stakeholders as well as joint Annual Work Plan between Sri Lanka, Ethiopia, and China;
- Ensure that all project partners understand project objectives and main activities and take ownership of the project;
- Plan TSSC Project board meetings to increase coordination between the two projects, including reporting modalities.

To ensure TSSC project scalability, policy impact and sustainability, the project will be implemented within broader engagement between UNDP Ethiopia and MOWIE. The management structure outlined in the diagram below represent the overall TSSC programmatic management structure for both Ethiopia and Sri Lanka projects. UNDP China will manage both projects under the same modalities to increase consistency and efficiency. A common Project Board will be convened to ensure overall project coordination and will be co-executed by UNDP China on behalf of MOFCOM, MOWIE and the Sri Lanka Ministry of Power and Renewable Energy.

The roles and responsibilities of both the overall management structure and national projects' management is described in detail below.



Project Board

This is the highest body for strategic guidance, oversight, and coordination of the project, and as such will guide project direction and advisory services. The Board is responsible for making, by consensus, management decisions for the TSSC project when guidance is required by respective Country Project Manager, including recommendation for UNDP approval of project work-plans and revisions. To ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value for money, fairness, integrity, transparency, and effective international competition. In case a consensus cannot be reached within the Board, final decision shall rest with the UNDP Country Office's concerned.

Responsibilities for joint review of project results and joint decision making will be maintained at the project board level. In addition, the project board's main functions include:

- Maintain general oversight of the Project;
- Ensure linkages to national development priorities in each of the represented countries and UNDP;
- Provide strategic guidance to Implementing Partners based on global trends and development priorities to improve the project;
- Provide strategic guidance to Implementing Partners based on South-South Cooperation trends and priorities to improve the project;
- To ensure appropriate and high-level buy-in to the project, the modality and its objectives
    - Ongoing tracking of the value of trilateral cooperation

    - to consolidate and disseminate experience on whether the approach is successful
- Provide feedback to national institutions and maintain ongoing dialogue with these;
- Provide strategic advice on how to scale-up the initiative;
- Approve any major changes in the project work plans;
- Ensure commitment of resources to support project implementation;
- Arbitrate any conflicts within the project.

In addition to typical roles in project management the four following roles[4]are highlighted:

- **Project Directors** *(also called Executive role)*: representing the project ownership to chair the group. This role typically held by Partner Governments or UNDP.
- **Development Partners** *(also called Supplier role)*: representing the interests of other parties which provide funding and/or technical expertise to the project. It includes representatives from the Chinese Implementing Partners (CAU and ACCA21) and UNDP.
- **Beneficiary Representatives**: representing the interests of those who will ultimately benefit from the project.
- **Project Assurance**: Project Assurance is the responsibility of each Project Board member; however, the role can be delegated. The project assurance role supports the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures that appropriate project management milestones are managed and completed. Project Assurance has to be independent of the Project Managers. UNDP Programme Officers, or M&E Officers, typically hold the Project Assurance role on behalf of UNDP.

Project reviews by this group are made at designated decision points during the implementation of the project, or as necessary when raised by the Project Coordinator. This group is consulted by the Project Coordinator for decisions when Project Manager's tolerances (normally in terms of time and budget) have been exceeded (flexibility). Based on the approved annual work plan (AWP), the Project Board may review and approve project quarterly plans when required and authorizes any major deviation from these agreed quarterly plans. It is the authority that signs off the completion of each quarterly plan as well as authorizes the start of the next quarterly plan. It ensures that required resources are committed and arbitrates on any conflicts within the project or negotiates a solution to any problems between the projects and external bodies.

The Project Board also plays a critical role in UNDP commissioned project evaluations by quality assuring the evaluation process and products, and using evaluations for performance improvement, accountability, and learning.

The Project Board will meet annually.

[4]https://intranet.undp.org/global/popp/ppm/Pages/Programme-and-Project-Management-Arrangements.aspx)

Project Coordinator

The Project Board appoints this role to coordinate, on behalf of the Project Board, day-to-day collaborative efforts at the overall project level. The project coordinator plays a key role in coordinating and incorporating the programmatic contributions of all partner countries and is responsible for:

- Compiling joint project reports, with specific inputs from each partner country,
- Facilitating communication and coordination among partners;
- Preparing for project board meetings
- Facilitating joint activities as needed; and
- Managing of Chinese implementing partners' technical and financial inputs to the TSSC project.
- The Project Coordinator is supported by the Coordinating Office, UNDP China.

Coordinating Office

UNDP China is appointed Coordinating Office to provide coordination services to the project. These services should include:

- Preparing joint project reports by consolidating information on results and resources;
- Facilitating management arrangements, for example by convening Project Board meetings;
- Facilitating communication and coordination between partners;
- Coordinating project evaluations and reviews, as needed;
- Facilitating joint activities, as needed;
- Acquiring, overseeing, and coordinating technical inputs from Chinese partners, as needed;
- Act as focal point for all interaction with MOFCOM.

The cost for services provided by a Coordinating Office should be covered by the project budget.

National South-South Task Force

A National South-South Task Force will be established to oversee project implementation and ensure coordination with all the project partners and related initiatives. The NSSTF will be aligned to the existing GEF Project Steering Committee and convene representatives from MOWIE, UNDP Ethiopia. Chinese Implementing Partners (i.e. UNDP China, ACCA21 and CAU) will be represented as needed. Its main responsibility will be to:

- Oversee implementation of the project;
- Review and approve annual work plans, deliverables, and budgets;
- Approve minor changes in project plans;
- Provide technical inputs and advices.
- Oversee project progress;
- Oversee the financial management and production of financial reports;
- Ensure that risks are being tracked;
- Provide direction and oversee revisions of annual work plans.

MOWIE will convene NSSTF coordination meetings twice a year. The Chinese Economic and Commercial Counsellor may sit in as an observer to ensure that MOFCOM interests are represented.

Project Management

The UNDP Country Office in Ethiopia

UNDP Ethiopia will provide project management support and coordination through assigned project manager.

UNDP manager will be responsible to:

- Oversee the project;
- Provide administrative and technical support;
- Control project finances including: (i) disbursement of funds to implementing partners, (ii) providing financial and audit services to the project; (iii) overseeing financial expenditures against project budgets approved by the TSSC Program Board; (iv) appointing independent financial auditors and evaluators; (v) collect, review and approve quarterly and annual reporting; and (vi) ensuring that all activities including procurement and financial services are carried out in strict compliance with UNDP procedures;
- Oversee interoffice communication and coordination;
- Manage international procurement;
- Monitor and evaluate the project as well as to ensure quality assurance.

The Project Management Unit

The Project Management Unit (PMU) within MOWIE will ensure the day to day management of the project. To increase coordination of project activities the PMU will be assisted by projects coordinators from GEF and TSSC projects. The project coordinators will ensure smooth project implementation and raise key concerns to be addressed as they arise and in a timely manner. Specific responsibilities will be to:

- Manage and coordinate project implementation in accordance with objectives, work plan and budget;
- Manage the day-to-day operations of the budget, including the management of financial and other records to facilitate audits of the project;
- Prepare work plans and associated budget;
- Plan and coordinate project activities and project-related meetings, incl. accounting and quarterly and annual progress reporting;
- Supervision of consultants and suppliers;
- Produce monitoring report for submission to UNDP.

Supporting South-South Cooperation and learning between Ethiopia and China

Project support will be provided by the Chinese government in form of a seconded staff. The staff member will support key project processes and ensure linkages to national institutions, private sector in both China and Ethiopia, and will function as a core member of the project management unit. The staff member shall have induction briefing and debriefing at the UNDP China office for the first two weeks and the last two weeks of the project support period.

The implementing partners will work closely together to implement joint activities as defined in the AWP. Such activities need joint planning, coordination and ongoing communication and close cooperation requires frequent communication via teleconference and in-person meetings. To the extent possible, project coordination shall be planned as an extension of planned study tours and exchanges to reduce direct costs. Other project coordination travel may be necessary.

The list of partners and their corresponding responsibilities:

Partner's Name	Type of Institution	Location	Responsibilities
UNDP Ethiopia	International Organization	Addis Ababa, Ethiopia	• Coordinates project and partners in Ethiopia • Project monitoring and evaluation, quality assurance • Member of the Project Board • Produce and deliver inputs for quarterly progress and final reports (in English)
UNDP China	International Organization	Beijing, China	• Overall project coordinator • Collaborate with UNDP Ethiopia office to ensure project objectives are met • Liaison with GoC/MOFCOM officials and other Chinese institutions (i.e., MOA-MOST) • Collect and consolidate inputs from partners to produce and deliver quarterly progress and final reports • Organize and conduct monitoring trips with MOFCOM and UNDP Ethiopia • Share quarterly and annual progress reports with MOFCOM (and translate into Chinese) • Produce, in collaboration with UNDP Ethiopia, final assessment of project for MOFCOM
Ministry of Agriculture	Government Organization	Beijing, China	• Technical partner • Assist with the identification and selection of Chinese technical experts or consultants, where applicable
ACCA21	Government Organization	Beijing, China	• Local Chinese project partner • Liaison with UNDP China • Partner for development of detailed project plans, including needs assessment, and project monitoring and evaluation • Assist with the identification and selection of Chinese technical experts or consultants, where applicable • Provide input and feedback to quarterly progress and final reports • Represented on the Project Board
China Agricultural University	Public Institution	Beijing China	• Local Chinese project partner • Liaison with UNDP China • Provide designing and training capabilities • Talent cultivation • Partner for development of detailed project plans, including needs assessment, and project monitoring and evaluation

			• Provide input and feedback to quarterly progress and final reports • Represented on the Project Board
Ethiopia Ministry of Water, Irrigation and Electricity	Government Organization	Addis Ababa, Ethiopia	• Local Financial and Technical project partner • Partner for development of detailed project plans, including needs assessment, and project monitoring and evaluation • Assist with the identification and selection of Chinese technical experts or consultants, where applicable • Provide feedback to quarterly progress and final reports • Represented on the Project Board
Energy studies Department	Government Organization	Addis Ababa, Ethiopia	• Local Technical project partner • Partner for development of detailed project plans, including needs assessment, and project monitoring and evaluation • Provide feedback to quarterly progress and final reports • Responsible for R&D activities
Alternative Energy Technology Development and Promotion Directorate	Government Organization	Addis Ababa, Ethiopia	• Local Technical project partner • Partner for development of detailed project plans, including needs assessment, and project monitoring and evaluation • Provide feedback to quarterly progress and final reports • Responsible for demonstration and implementation activities

VI. Monitoring Framework and Evaluation

An inception phase will bring all project partners from Sri Lanka, Ethiopia, and China together to:

- Coordinate with MOFCOM and UNDP Ethiopia to confirm whether a final evaluation will be required and plan a budget line accordingly.
- Define a detailed monitoring and evaluation plan that considers and proposes coordination modalities for reporting on national components activities as well as overall project progress, as described below.

The project is composed of one national and a common component with Sri Lanka. The monitoring will therefore request coordination between reporting project progress nationally and globally.

- A detailed Monitoring Plan will be prepared, to capture all outcome 1 activities (i.e. national component) the project carries out in Ethiopia.
- A Joint Monitoring Plan will be prepared to capture all activities the project carries out in China, Sri Lanka and Ethiopia. Ethiopia TSSC project manager in collaboration with UNDP China will prepare an overall Joint Monitoring Plan for all activities carried out under outcome 2 and determine how to integrate the monitoring of national components.

In accordance with the programming policies and procedures outlined in the UNDP POPP, the project will be monitored through the following:

Inception Report:

The inception report should address agreements reached during the stakeholders' workshop including:

- Outline of programme and project management procedures and set-up including elaboration of the project administration, communication and coordination manual;
- Revised Annual Work Plan between Ethiopia and Chinese stakeholders as well as joint Annual Work Plan between Sri Lanka, Ethiopia and China;
- Schedule of TSSC Programme Project board meetings;
- Monitoring and Joint monitoring plans, as well as evaluation plan if deemed necessary.

Quarterly Report:

Project managers will provide to UNDP a quarterly report outlining project progress and undertaken monitoring activities including site visits. A financial report will also be generated. The risks will be monitored and the risk log updated accordingly.

Annual Report:

A project review shall be conducted annually to assess the performance of the project and appraise the Three-Year Work Plan (TYWP). In the last year, this review will be a final assessment. The project managers will report annual assessment findings in an Annual Review Report following UNDP's reporting to be shared with UNDP and presented to the TSSC Project Board.

VII. Legal Context

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of China and UNDP, signed on 29 June 1979, and between the Government of Ethiopia and UNDP, signed on 6 February 1981.

Consistent with the Article III of the SBAA, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:

a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;

b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document [and the Project Cooperation Agreement between UNDP and the Implementing Partner].

The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/Docs/sc/committees/1267/1267ListEng.htm.This provision must be included in all sub-contracts or sub-agreements entered into under/further to this Project Document.

VIII. ANNEXES

Risk Analysis

Project Title: Biogas and Solar trilateral cooperation - Transitioning to Sustainable Energy uses in the Agro-Industry. Ethiopia Project	**Award ID:**	**Date:**

#	Description	Type	Impact & Probability	Countermeasures / Mngt response
1	Political instability in pilot project country	Political	Delays in Project implementation or project postponed Probability = 3 Impact = 5	Project partners cannot mitigate this risk.
2	Insufficient maintenance of equipment provided	Operational	Insufficient maintenance would lead to premature deterioration and break down of equipment leading to non-sustainability of project interventions Probability = 3 Impact = 5	On-site training is planned targeting local administration, local association and end-users who will run the demonstration sites. The project will also provide extensive training on operation, and safety to build a pool of experts. Finally, the project will provide each demonstration sites with manual and information material.
3	Inappropriate selection of technologies/ providers	Operational	Inappropriate selection of technologies and providers would lead to rejection by users and non-performance of demonstrations sites resulting in project failure Probability = 2 Impact = 5	This is a major component of the project. Preliminary activities have been defined around technology selection for a specific use to ensure adapted technology. Furthermore, UNDP China and UNDP Ethiopia will closely monitor and QA activities.
4	Inappropriate selection of pilot	Operational	Inappropriate selection of	This is a major component of the project. Preliminary

	demonstration sites and end-users		demonstration sites would lead to rejection by users and non-performance of demonstrations sites resulting in project failure Probability = 2 Impact = 5	activities have been planned to ensure sites are selected based on consistent criteria. Experts will be define these criteria and feasibility studies will be conducted. Furthermore, UNDP china and UNDP Ethiopia will closely monitor and QA activities.
5	Differences among member countries in the Project Board	Political	Changes in the bilateral relations amongst China and Ethiopia may affect mobility of goods and people. Probability = 1 Impact = 5	Project partners cannot mitigate this risk, but can ensure/increase coordination with the Chinese Embassy in Addis Ababa
6	Socio-environmental impact for the demonstration project not properly assessed	Operational	If not properly assessed, there is a likelihood of affecting the ecosystems and livelihood. Probability = 2 Impact = 4	The project requires a socio-environmental impact assessment. The activity will be implemented and supervised in collaboration between MOWIE and UNDP
7	The South-South Cooperation Centre is not established under Ghana-Zambia projects.	Operational Political	Without the centre established, the project would be slowed down and face challenges to scale up the initiative. Exchange of knowledge may be lost. Probability = 2 Impact = 4	This Centre is in line with ACCA21's organizational vision and is being set up as the new target to achieve an effective south-south technology transfer. Most importantly, this Centre is fully resourced by Ghana-Zambia project, and additional resources will be provided under this project to institutionalize the role of the centre.
8	Local stakeholders in China do not want to engage in project led initiatives leading to limited technology and Know-how transfer	Operational	Chinese identified partners might decide not to support and be involved in project interventions leading to delays in project implementation, incl. technology	An initial stakeholder meeting with all partners will be held in Ethiopia. ACCA21 is already engaged as a South-South partner and has a direct interest in embedding additional projects to institutionalize the South-South

			transfer and exchange of knowledge. Probability = 3 Impact = 3	Cooperation Centre. The MOA is the leading ministry in biogas research and technology transfer.
9	Lack of Coordination between implementing partners	Operational	Poor coordination would lead to delays in project implementation Probability = 3 Impact = 3	A communication and coordination of activity strategy will be defined at the start of the project to ensure coordination and facilitate communication. UNDP will supervise the development of these strategies to ensure consistency and lessons learnt are used from past projects.
10	Change in UNDP's approach to South-South cooperation.	Organizational Strategic	Delays in Project implementation Probability = 1 Impact = 3	MoU signed between China and UNDP. South-South Cooperation aligned with UNDAF
11	Budgeting delays in project beneficiary country	Organizational Strategic	Delays in Project implementation Probability = 1 Impact = 3	The project is designed as incremental support to existing projects in Ethiopia, for example GEF project on rural energy access. Incremental activities to existing projects will be co-financed through these projects.

Other Partner Organizations

The Horn of Africa Regional Environment Centre: manages a network of 40 higher learning institutions and research centres, civil society and community-based organizations in the Horn of Africa. It focuses on environmental concerns and sustainable development options within the Horn of Africa. It facilitates, strengthens and advocates for initiatives related to environmental conservation and natural resource management. The network is built upon three main partnership programs: The Climate Resilience Partnership Program, the Social Ecological Resilience Partnership Program and the Environmental Governance and Education Partnership Program.

Ethiopian Ministry of Education

Ethiopian Ministry of Science and Technology

World Bank Group: The World Bank Group support for energy sector activities in Ethiopia including financing of major power sector projects (e.g. universal electricity access program), support for assessment of renewable energy resources (national wind resource mapping), promotion of energy efficiency (CFL distribution project), and investment facility to build capacity of clean energy SMEs.

WBG engagement in the off-grid (and small scale renewable energy efficiency) area is focused on development the capacity (financing) of private sector SMEs delivery sustainable energy services. WBG is running the Lighting Ethiopia program which promotes WBG approved solar home systems for off-grid consumers in Ethiopia. The WB/IFC has also provided credit line amounting to US$20 million as working capital loans for solar SMEs (import and distribution) and micro finance banks. Additional financing of US$20 million is recently approved to expand this program. WBG estimates that 0.8 million solar home systems have been distributed so far due to the program. The WBG is now promoting risk mitigating actions to address barriers of access from borrowers (collateral requirements for enterprises, others).

USAID Power Africa: Power Africa (under the USAID) is a multi-partner initiative launched in 2013 with the objective of increasing access to electricity in sub-Saharan Africa. Power Africa promotes the utilization of Africa's renewable energy resources including wind, solar, geothermal, hydropower and biomass.

In Ethiopia Power Africa provides strategic support to the energy sector mainly for IPP engagement in the power sector including development and revision of laws and regulations, developing the grid code, assistance in planning and O&M of power facilities, and development regulatory capacity of the Ethiopian Energy Authority including tariff setting.

Power Africa also provides small grants (maximum of US$ 100,000 per project) to promote off-grid electrification and other technologies. It has provided 4 such grants in the past three years. past recipients of these grant included wind and solar photovoltaic powered micro grid for off-grid applications.

SNV: National Biogas Program of Ethiopia (NBPE) – SNV provides technical support/advice to the NBPE. The NBPE has installed about 14,000 domestic biogas units (4 to 10m3 capacity units) since 2009. The **National Biogas Program of Ethiopia**: has applied and is expecting additional financing amounting to Euro 20 million from the EU Delegation in Ethiopia to expand the program in the next five years (to install 100,000 more domestic biogas units)

GIZ: The GIZ energy program has been running in Ethiopia since 1999. The main focus areas for GIZ have been improved domestic biomass cook stove, solar systems for institutions (health centres) and more recently micro hydropower for community electrification.

BIOMA: The Chengdu Biogas Scientific Research Institute (BIOMA) is the lead research institution in the development of the theory and process of biogas fermentation, including advanced research on micro-biology. BIOMA has been a major contributor to the development of rural environmental engineering, safe organic wastes disposal in rural areas and containment of water pollution. It also operates an international training and research centre in Chengdu, Sichuan Province. The centre is already training biogas experts from Southern countries through bilateral agreements managed by the Chinese Embassy locally and funded by MOFCOM. The project will provide support in identification and application of potential candidates to the BIOMA program. Experts from BIOMA may provide technical inputs to the project as requested.

The Department of Environment Protection and Rural Energy (DEPRE) under the China Ministry of Agriculture: Responsible for the execution of biogas programs, the DEPRE may provide additional technical support through their network of biogas experts and private companies. The project may draw upon this pool of experts and MOA past experience to leverage support in business model formulation and capacity building to reinforce the development of the energy service industry and technology dissemination modalities (incl. financial aspects).

List of Stakeholders consulted during the project formulation mission

Organization	Department	Contact
UNDP China		Devanand Ramiah, Deputy Country Director
	Energy & Environment	Zhang Weidong, Programme Manager
	South-South Cooperation Policy and Partnership Unit	Niels Vestergaard Knudsen, Team Leader & Assistant Country Director
		Teng Yue, Government Relations Officer
China Ministry of Agriculture Rural Energy & Environment Agency	Division of International Cooperation	Prof.Quanhui Wang, Director
	Division of Renewable Energy	Suu Liying, Senior Engineer
The administrative Centre for China Agenda 21 (AAC21)	Global Environmental Affairs	Dr.Jiutian Zhang, Director
		Zhang Xian, Project Director
China Agricultural University	Key Laboratory of Clean Production and Utilization of Renewable Energy, Ministry of Agriculture (CPURE)	Dr.Renjie Dong, Executive Deputy Director
	Biomass Engineering Center	Dr.Yuguang Zhou
BIOMA		Taipien, Director General
	Training and Information Research Center	Libin Wu, Program Officer
UNDP Ethiopia		Kidanua Abera, Energy & Low Carbon Development Analyst
		Wubua Mekonnen,
		Samuel Bwalya, Director
Ministry of Water, Irrigation and Electricity	Directorate for Energy Studies and Follow-up	Sahele Tamiru, Director
	Directorate for Alternative Energy Technology Development and Promotion	Asress Wolde-Giorgis, Director
Horn of Africa Regional Environment Center and Network		Araya Asfaw, Director
		FitsumberhanTsegaye, Coordinator for energy program
Ministry of Science and Technology	Technology Transfer and Development	TassisaKaba, Director General
	Science and Technology Universities and Institution	Shumete Gizaw, Director General
Climate Innovation Center (CIC)		Tehut Tesfaye, CEO

Solar Technologies Manufacturing (STM)		Samson Tsegaye, CEO
Lydetco, Solar Energy products supplier		DerejeWalelign, Manager of Lydetco, and President of Solar Energy Development Association
SNV Ethiopia		Aster Haile, Renewable Energy Advisor
		Mekonen Mekuria, Bio-Slurry Value Chain Advisor
		Worku Behonegne, Country Director
Addis Ababa University	Addis Ababa Institute of Technology	Fitsum Saleh, Acting Head, Center of Energy Technology
	Natural Science Faculty	BilhatLeta, PhD researcher on biogas
GIZ	Bioenergy Section of Energizing Development Project	Berhanu Negash, Head
	Electrification Section of Energizing Development Project	Ralf Elben, Head
Selam Children Village		Aschalew, Manager

TORs Trilateral South-South Cooperation Project Coordinator

General:

The Trilateral South-South Cooperation Project Coordinator will oversee the harmonized implementation of the Trilateral South-South Cooperation project activities and ensure overall project management, including financial management, on a full-time basis.

The Trilateral South-South Cooperation Project Coordinator will work closely with all implementing partners, UNDP Ethiopia, UNDP China, the Ministry of Water, Irrigation and Electricity (MOWIE), Regional Energy Bureaux (to be identified during the inception phase), and other local partners.

Duties and Responsibilities

The Project Coordinator will report to MOWIE (and specifically to the Energy Studies and Follow-Up Directorate under MOWIE) and UNDP Ethiopia and will be responsible for timely project implementation. The Project Coordinator will manage the project on a day-to-day basis on behalf of MOWIE, within the Government and UNDP procedures; with policy guidance from the Project Board. The Project Coordinator will be responsible for activities coordination; financial management and disbursements, with accountability to MOWIE and UNDP. The Project Coordinator will work closely with other responsible parties, the UNDP China Country Office, and other Project Implementing Partners.

Coordination of Project Activities:

- Coordinate implementing partners in carrying out their tasks to ensure timely project implementation;
- Maintain close linkages with relevant sectoral government agencies (e.g. Ministries and Bureaus of Health and Education), UNDP, NGOs, civil society, international organizations and implementing partners of the project;
- Recruit and supervise technical and training experts as required for implementation of the project, including preparation and approval of ToRs;
- Coordinate recruited experts in carrying out their tasks in a timely, project objective oriented and cost-effective manner;
- In collaboration with implementing partners, prepare quarterly and annual work plans and budgets;
- Compile, summarize, record and maintain information on project activities and its financial data;
- Prepare, update and maintain project P.R. materials such as brochures;
- Facilitate and coordinate with project partner compilation and circulation of project literature to all interested parties.

Support Implementing Partner in Event Management – Organizing workshops and meetings:

- Assist in the organization of workshops, seminars, and round table meetings including preparation of workshop materials and tools;
- Prepare budget and draft requests for disbursements of funds for project staff participation to workshops, meetings and other events.

Provision of Progress and Monitoring Reports:

- Coordinate inputs into all project reports as required (including Annual Project Reports, Inception Report, Quarterly Reports and the Terminal Report);
- Prepare project progress reports (technical and financial) for comments and approval by the Implementing Partner and UNDP Country Office;
- Support implementing partners for the monitoring of project activities and facilitate monitoring process to ensure timely and quality delivery of monitoring reports;
- Coordinate with UNDP China to ensure project progresses and challenges are all captures in all joint reports.

Project Management:

- Ensure all general administrative matters of the project are managed effectively.
- Preparation of required budget revisions and drafting requests for disbursements of funds under the project;
- Assisting in the compilation of required documentation for project audits;
- Support MOWIE in the implementation of the audit recommendations in a systematic manner;
- Supporting the monitoring of project activities by previewing a variety of records, including control plans, progress reports, project inputs, budgets and financial expenditures;
- Undertaking visits to project sites and preparing reports on them
- Documenting of all project meetings and other activities as required.
- Raise key concerns to be addressed as they arise and in a timely manner;
- Provide additional support as requested and reasonable by the implementing partners.

Qualifications and experience:

- A minimum of Master's degree or equivalent in renewable energy, engineering natural sciences, environmental management, institutional / organizational development, or related fields.
- A minimum of 5 years working experience in areas related to at least one or two of the project thematic areas (renewable energy, energy services delivery, climate change mitigation and adaptation, etc.);
- Previous experience in the management of large, complex projects;
- Demonstrated facilitation and coordination skills, ability to manage multi-country and multi–stakeholder teams;
- Practical experience in process-oriented and strong ability to work closely with diverse teams, conflict resolution and negotiation skills;
- Fluency in English.

Duty Station:

The Project Manager will work at the Ministry of Water, Irrigation and Electricity offices in Addis Ababa and should be willing to travel outside Addis Ababa as needed.